Answer key: Lesson 1, Half and Quarter

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the three shapes. Is half of each shape coloured? Circle Yes or No for each shape.

Answer: A yes; B no; C yes

Shape A has 2 equal parts and 1 is coloured, so half is coloured: Yes. Shape B has 4 equal parts and 1 is coloured, so a quarter is coloured, not a half: No. Shape C has 2 equal parts and 1 is coloured, so half is coloured: Yes.

Watch for: Saying Yes for any coloured part.; Not counting the parts: a quarter is coloured in shape B.

S2. The circle is cut into equal parts. Fill in the blanks.

Answer: 4 equal parts; quarter

The circle has 4 equal parts. One of 4 equal parts is a quarter.

Watch for: Counting the coloured part only and writing 1.; Writing half: a half is one of 2 equal parts.

S3. Colour the shapes.

Answer: any one part of the square; any one part of the circle

The square has 2 equal parts: colour 1 of them for a half. The circle has 4 equal parts: colour 1 of them for a quarter.

Watch for: Colouring 2 parts of the circle.; Colouring both parts of the square.

S4. Fill in the blanks.

Answer: 2; 4

The circle has 2 halves, and 2 halves make 1 whole. The square has 4 quarters, and 4 quarters make 1 whole.

Watch for: Writing 1, the number of wholes.; Mixing up the two pictures.

S5. Look at the four shapes. Write the number of the shape.

Answer: 2; 3

Shape 2 is a half circle: half of a circle, with 1 straight side. Shape 3 is a quarter circle: a quarter of a circle, with 2 straight sides that make a corner.

Watch for: Choosing the whole circle for the half circle.; Mixing up shapes 2 and 3.

S6. Use your ruler. Draw straight lines on the squares.

Answer: square A: 1 line through the middle; square B: 2 lines through the middle that cross

For square A, 1 line through the middle (across, down or corner to corner) makes 2 equal parts. For square B, 2 lines through the middle that cross make 4 equal parts.

Watch for: Drawing a line that is not through the middle, so the parts do not match.; Drawing 4 lines instead of 2.

Think harder (page 3)

T1. Cut each square into 2 equal parts. Use your ruler. Make a different cut each time.

Answer: any three different lines through the middle: down, across, corner to corner

Down the middle, across the middle and corner to corner each make 2 equal parts. The parts match when we fold, so each part is a half. A half of a square can be a rectangle or a triangle.

Watch for: Drawing a line that is not through the middle.; Drawing the same cut on all three squares.

T2. The rectangle has 8 small squares. Fill in the blanks.

Answer: 4; 2

A half is 2 equal parts of 8 squares: 4 and 4, so a half is 4 squares. A quarter is 4 equal parts: 2, 2, 2 and 2, so a quarter is 2 squares.

Watch for: Writing 8 for the half.; Writing 4 for the quarter.

T3. Tick True or False for each statement.

Answer: true; false; true; false

A half of a square can be a triangle: True (fold corner to corner). Any 2 parts of a shape are halves: False, they must be equal. 2 quarters of a circle make 1 half of the circle: True (we laid 2 quarters on 1 half). A half circle is the same size as a quarter circle: False, a half circle is bigger.

Watch for: Saying True for any 2 parts of a shape.; Thinking a half cannot be a triangle.

T4. Mei folds a paper circle in half again and again. Fill in the table.

Answer: 2, 4, 8

1 fold makes 2 equal parts. A 2nd fold cuts each part in 2: 4 parts. A 3rd fold cuts each of the 4 parts in 2: 8 parts. The parts double with each fold.

Watch for: Adding 1 part for each fold (writing 3 or 5).; Writing 6 for 3 folds.

Go further (page 4)

G1. Same Half, Different Shape. Each rectangle has 8 small squares. A half of each is coloured. Look at the three rectangles.

Answer: 4, 4, 4; Yes; No

Each coloured part has 4 small squares, and the white part has 4 as well, so each coloured part is a half. The coloured parts have different shapes: a long bar, a block of 4 and a bent shape. Equal parts do not have to be the same shape.

Watch for: Thinking halves must look the same.; Counting the white squares and writing 4 for only one rectangle.

G2. Guess the shape. Read the clues and circle the shape.

Answer: half circle; triangle

Riddle a: 1 straight side and 1 curved side, and 2 of me make a circle: a half circle. Riddle b: cut from a square, 3 corners, and 2 of me make the square: a triangle.

Watch for: Choosing the quarter circle for riddle a: 4 of those make a circle, not 2.; Choosing the rectangle for riddle b.

G3. The strip is a quarter of a rectangle. Fill in the blanks.

Answer: 12; 6

The strip has 3 small squares and it is 1 quarter. 4 quarters make the whole: 3, 6, 9, 12. The whole has 12 small squares. A half is 2 quarters: 6 small squares.

Watch for: Writing 4 or 7 (adding instead of making 4 equal groups).; Writing 12 for the half.

Investigation (page 5): Fold twice

First: 2 parts. Then at least 2 different ways, and 3 are easy to find: two lines down and across (4 small squares), two corner-to-corner lines (4 triangles), or 3 parallel lines (4 strips). Each way gives 4 equal parts, but the parts have different shapes. Fold the square to check that the parts match.

Look for: A child who folds a paper square to test a way, who says 4 equal parts can look different (squares, triangles, strips), and who checks by laying the parts on top of each other. Credit any clear reason. Do not mark a child down for stopping at one way.

Exit ticket (page 5)

E1. The square is cut into equal parts. One part is coloured. Is the coloured part a half or a quarter? Circle one.

Answer: quarter

The square has 4 equal parts, so one part is a quarter.

Watch for: Saying half for any coloured part.; Not counting the parts.

E2. How many halves make 1 whole? Write the number.

Answer: 2

2 equal parts make 1 whole, and each part is a half.

Watch for: Writing 1.; Writing 4, the quarters.

E3. A square is cut into 2 parts that are not the same size. Are they halves? Circle Yes or No.

Answer: No

Halves must be equal. Parts that are not the same size are not halves.

Watch for: Saying Yes because there are 2 parts.; Thinking the bigger part is a half.

Sneak-peek puzzles (not in the workbook)

1. 2
2. 4. The first fold makes 2 equal parts. The second fold cuts each of those parts into 2 equal parts, so 2 and 2 make 4.